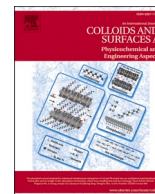




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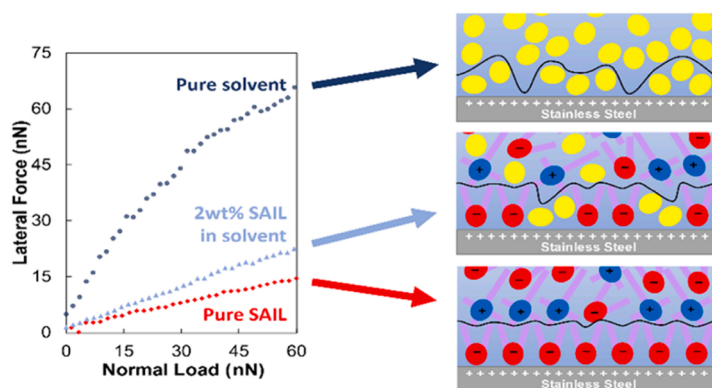
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Surface-active ionic liquids as lubricant additives to hexadecane and diethyl succinate

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GRAPHICAL ABSTRACT



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ABSTRACT

The effectiveness of surface-active ionic liquids (SAILs) as lubricant additives in both polar and apolar base liquids has been investigated on stainless steel surfaces at both macroscale and nanoscale. The SAILs studied are composed of trihexyltetradecylphosphonium ($[P_{6,6,6,14}]^+$) cations paired with either dioctyl sulfosuccinate ($[AOT]^-$) or dodecyl sulphate ($[DS]^-$) anions. The polar and apolar base liquids used to dilute the SAILs are diethyl succinate ($(CH_2CO_2Et)_2$) and hexadecane (HD), respectively. The friction coefficients of pure $[P_{6,6,6,14}] [AOT]$ and $[P_{6,6,6,14}] [DS]$ are up to $\sim 80\%$ lower than those of $(CH_2CO_2Et)_2$ and HD. Remarkably, diluted solutions containing as little as 5 wt% SAIL in either $(CH_2CO_2Et)_2$ or HD exhibit lubricity comparable to that of pure SAILs, revealing their excellent potential as lubricant additives. The high lubricity of the SAIL solutions in both $(CH_2CO_2Et)_2$ and HD is attributed to the adsorption of anions onto the positively charged stainless steel surface, resulting in the formation of a robust boundary layer that reduces surface contact and facilitates smooth sliding. These insights aid the development of advanced, cost-effective lubricant additives, with potential applications across various industrial sectors.

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1. Introduction

Ionic liquids (ILs) are pure salts with low melting temperatures, often referred to as “designer solvents” due to the ability to fine-tune their properties by modifying the molecular structures of their constituent cations or anions [1–4]. ILs and other similar highly concentrated electrolytes are known for their remarkable properties, including high thermal and electrical conductivity, low vapor pressure, and low flammability [5,6]. These characteristics have incited research interest in a wide range of applications, such as lubrication [6], sensors [7], solar cells [8], electrolytes [9], biomass processing [10], and coal pre-treatment, where IL ions disrupt hydrogen bonding and π - π stacking [11], leading to changes in pore geometry [12,13] and molecular structure [14].

Despite the remarkable properties of ILs, their widespread application is often hindered by their high cost. One strategy to address this challenge is to develop cost-effective materials with similar properties, such as deep eutectic solvents (DESs) [15–21]. Another approach is to enhance key attributes of ILs, as exemplified by surface-active ionic liquids (SAILs), which exhibit more well-defined nanostructures compared to traditional ILs. SAILs comprise of at least one ion with a surfactant-like structure, usually the anion, featuring a charged polar centre and one or more long apolar alkyl chains [22,23]. The counterion can be a conventional IL ion or it can also possess a surfactant-like structure, resulting in a catanionic SAIL. For instance, SAILs have been prepared using surfactant-like anions including dioctyl sulfosuccinate ([AOT][−]) [23] and dodecyl sulfate ([DS][−]) [22] in combination with surfactant-like cations such as trihexyltetradecylphosphonium ([P_{6,6,6,14}]⁺) [24] or conventional cations like 1-butyl-3-methylimidazolium ([Bmim]⁺) [25]. Compared to conventional ILs, SAILs feature defined amphiphilic properties caused by polar heads and long apolar alkyl chains [22,23,26,27].

SAILs often exhibit strong amphiphilic nanostructures [2,24,26–30], which arise from electrostatic attractions between the charged groups of the SAIL ions. These interactions lead to the formation of charged domains, from which the long alkyl chains are solvophobically excluded into separate apolar domains [31]. The polar and apolar domains percolate through the bulk, creating a sponge-like structure [2,28,29]. This amphiphilic nanostructure enables various useful properties including the ability to dissolve both polar and apolar compounds and the formation of robust interfacial structures that result in high capacitance [26]. SAILs have also been explored for applications in other electrochemical devices [26,27,32], gas absorption and capture [26, 33–35], and drug delivery [36], in addition to their use as lubricants [24], which is the primary focus of this article.

Lubricants are substances, typically liquids, that form a barrier between moving surfaces to reduce friction [37]. In the hydrodynamic lubrication regime, the sliding surfaces are physically separated by the liquid, and energy dissipation (i.e., friction) is determined by the hydrodynamic pressure arising from the lubricant's viscosity [38]. In the mixed lubrication regime, intermittent contact occurs between the surfaces, resulting in friction in addition to the hydrodynamic pressure [38]. In the boundary lubrication regime, the bulk of the liquid has been squeezed out, leading to more frequent surface contact, and lubrication is attributed to the boundary layers adsorbed onto the sliding surfaces [38]. **Consequently, friction in the boundary regime is independent of the viscosity of the lubricant. Lubrication, or friction reduction, arises from energy dissipation, which in the boundary regime is typically proportional to the roughness of the shear plane parallel to the boundary layer [39,40].** The two key energy dissipation mechanisms for AFM *in situ* measurements of ILs are the compression, rotation, and oscillation of the ions enriched in the boundary layer, and the expulsion of ion layers [39,40]. The amount of energy dissipation is, respectively, mainly controlled by the ion structure, conformation and packing density, and the lateral cohesive interactions between ions in the boundary later and the innermost near-surface layer [39,40].

Conventional lubricants typically adsorb weakly onto surfaces and are easily expelled under high contact pressure [37], limiting their effectiveness in the boundary regime. In contrast, ILs often exhibit excellent boundary lubrication properties [41–43] due to strong electrostatic interactions between the IL ions and charged surfaces. ILs form robust boundary layers enriched in counter-ions that are strongly adsorbed onto the surface. These boundary layers are capped by a well-defined near-surface layer, which sometimes takes the form of micelle-like aggregates [44]. **The long alkyl chains of SAILs (usually ≥ 12 carbons) often form better-defined interfacial structures compared to conventional ILs (alkyl chains typically ≤ 8 carbons), which enhances their lubricity, particularly in the boundary regime [24,30].** Although pure SAILs may be too expensive for use as a lubricant in most applications, except for niche purposes where the cost-benefit is favourable, they can still be used as lubricant additives in inexpensive base liquids.

Lubricant additives are substances that are added in small quantities to base liquids, such as base oils or esters, to improve boundary layer adsorption or tribofilm formation, thereby enhancing resistance to friction and wear under high pressure [5]. Typical lubricant additives include **zinc dialkyldithiophosphates (ZDDPs) and nanoparticles**. The morphology and properties of the tribofilms formed by ZDDPs on stainless steel have been studied with various techniques. Consistent film thickness and robustness have been found as normal load increases [45–47]. The efficacies of ILs and ZDDPs as lubricant additives have been investigated and compared when added to an apolar base oil, polyalphaolefin (PAO). The results show that ILs can form a more robust tribofilm than ZDDPs [48]. **Titanium dioxide nanoparticles** have also been investigated as lubricant additives due to their excellent ability to form protective tribofilms. Further studies have shown that the concentration and size of the nanoparticles have a critical role in their effectiveness as lubricant additives [49–51].

Currently, limited studies have explored the potential of ILs as additives in water-based lubricants for stainless steel [49, 52–56]. For instance, the addition of tributylmethylphosphonium dimethylphosphate to a water-glycol base lubricant resulted in reduced friction compared to the base lubricant alone. However, its performance was inferior to that of the reference additive, dodecanoic acid [55]. In contrast, 1-butyl-1-methylpyrrolidinium tris(pentafluoroethyl)trifluorophosphate demonstrated remarkable friction reduction capabilities, surpassing even dodecanoic acid. This IL formed a robust tribofilm at the interface, contributing to its enhanced lubrication properties. Nevertheless, the potential decomposition of the IL and the associated risk of fluorine release pose significant challenges for its practical application [55]. A series of water-soluble protic IL additives exhibited scarce corrosive attack to steel in water environment and showed extremely high stability as well as lubricity due to self-assembly of IL ions into bilayers at the interface [56].

Phosphonium-based ILs have been found to be miscible in a variety of base oils, including HD, linoleic acid, pentaerythritol ester, and trimethylolpropane ester [5, 57–59], indicating their potential as lubricant additives. Studies have examined phosphonium-based ILs as lubricant additives in base apolar oils [60,61] or polar esters [57,62,63] on steel surfaces, reporting similar improvements in lubrication. IL cations and anions adsorbed onto metal surfaces can react to form a tribofilm that protects the surface [64–68], with phosphonium ions showing a particularly strong tendency to form protective tribofilms [69–72]. Oil-miscible phosphonium-based ILs have been investigated as lubricant additives in hexadecane (HD) for silica and alumina surfaces [59,73]. Even at low concentrations of the IL in HD, friction was similar to that of the pure IL and significantly lower than that of the pure oil at both the macroscale and the nanoscale.

For IL lubrication, the composition and orientation of ions in the boundary layer are crucial, particularly in the boundary lubrication regime [39, 74–79]. Both of these factors can be tuned by applying a potential to a conductive surface [80–83]. This integration of tribology and electronics is known as tribotronics [84]. When a positive potential

is applied, anions are attracted to the surface while cations are repelled, whereas the opposite occurs when a negative potential is applied. This allows external control over the composition of the boundary layer and, to some extent, the preferred orientation of ions [80,85]. Tribotronics has previously been demonstrated for conventional ILs dissolved in base oils [79]. For instance, in the case of trihexyl(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl) phosphinate, the cation-rich boundary layer formed at negative potentials exhibited better lubrication properties than the anion-rich layer formed at positive potentials, even at concentrations as low as 5 mol%.

Recent atomic force microscopy (AFM) studies have shown that the SAIL $[P_{6,6,6,14}]^+ [AOT]^-$ forms well-defined interfacial structures and shows excellent lubricity when applied to gold surfaces [24,30]. However, despite their excellent solubility in diverse liquids and pronounced surface activity, typical SAILs have not been widely investigated as lubricant additives [86]. To address this knowledge gap, in this work, we investigate the effectiveness of two SAILs, $[P_{6,6,6,14}]^+ [AOT]^-$ and $[P_{6,6,6,14}]^+ [DS]^-$, as lubricant additives in a polar base liquid, diethyl succinate $((CH_2CO_2Et)_2)$, and an apolar base liquid, HD, on stainless steel surfaces at both macroscale and nanoscale [52].

The interfacial nanostructure and lubricity were studied by macrotribology measurements performed using a rheometer with a 3-ball-on-plate geometry and nanotribology measurements and force-separation profiles by using AFM. The presence of phosphorus in these SAILs could be seen as a potential drawback for their potential applications due to environmental regulations. While the presence of phosphorus in SAILs may raise environmental concerns, phosphonium-based ILs offer superior thermal and base stability [87,88] compared to conventional nitrogen-based ILs. Moreover, the small quantities of SAILs used as lubricant additives minimize their environmental impact, making them a promising choice for advanced lubrication systems. The findings from this work will assist further development of high performance and cost-effective lubricant additives, with potential applications across various industrial sectors, such as aerospace [89–91] and automotives [92–95].

2. Materials and methods

2.1. Chemicals and materials

Trihexyltetradecylphosphonium chloride $([P_{6,6,6,14}]^+ [Cl]^-)$, >95 % was provided by IoLiTec (Ionic Liquids Technologies GmbH) and used as received. Sodium dioctyl sulfosuccinate $(Na [AOT]^-)$, sodium dodecyl sulfate (SDS), dichloromethane (DCM), and silver nitrate were purchased from Sigma-Aldrich and used as received. Stainless steel sheets (Grade 304) were purchased from Total Laser & Plasma Cutting Services in Bayswater, Western Australia. It was selected as the substrate material in this study due to its widespread use across a broad range of applications [96]. Stainless steel has a point of zero charge at pH 8.5 [97]. Since the pH of SAILs is typically near neutral [98], stainless steel surfaces are positively charged in SAIL-based systems. To ensure consistent surface conditions, the stainless steel sheets were thoroughly cleaned with acetone and ethanol in a sonicator before each experiment.

2.2. Synthesis and characterization

20 mmol of $[P_{6,6,6,14}]^+ [Cl]^-$ and 20 mmol of either $Na [AOT]^-$ or SDS were mixed and dissolved in DCM and stirred at room temperature for 4 h. The precipitate formed was removed by vacuum filtration. Chloride was removed from the organic phase by washing with water, which was confirmed by the absence of silver precipitate formation when mixed with silver nitrate in aqueous solution. A rotary evaporator was used to remove DCM, and a Schlenk line was used to remove water. Both samples were measured with Karl-Fischer titration measurements and no water was detected. The chemical structures of the synthesized SAILs were confirmed by 1H NMR (Figure S1 and Figure S2).

2.3. Miscibility tests

A series of samples with varying concentrations of $[P_{6,6,6,14}]^+ [AOT]^-$ or $[P_{6,6,6,14}]^+ [DS]^-$ in HD or $(CH_2CO_2Et)_2$ were prepared. After stirring at room temperature for 30 min and settling for three days, the miscibility was determined by observing which samples remained homogeneous without phase separation.

2.4. Viscosity tests

Viscosity tests were conducted using a DHR-3 rheometer (TA Instruments) with an attached Peltier plate. The geometry used was a 2° stainless steel plate with a diameter of 40 mm. For viscosity vs shear rate tests, the shear ramp swept from $1\ s^{-1}$ to $5000\ s^{-1}$ linearly over a 100 s period at a fixed temperature of 25 °C (298 K). All runs were performed in duplicate.

2.5. Macroscale friction tests

Friction at the macroscale was conducted using a DHR-3 rheometer with a 3-ball on plate geometry (TA Instruments). Each test was run at $15\ mm\ s^{-1}$ with $2.0 \pm 0.1\ N$ of normal load, a sliding distance of 4.5 m, and temperature at 25 °C (298 K). $15\ mm\ s^{-1}$ with $2.0 \pm 0.1\ N$ of normal load were chosen to be consistent with previous studies [38,99]. Data points are the averaged results of runs conducted in duplicate. In cases where the experimental error between the two repeats exceeded 10 %, a third repeat was conducted to ensure the reliability and reproducibility of the data. 40x40 mm stainless steel sheets were cleaned via sonication in ethanol for 30 min to remove any contaminants before use. The surface roughness of the stainless steel sheets, measured using AFM, was $0.9 \pm 0.3\ nm$ at a scan scale of 500 nm by 500 nm [53]. Stainless steel balls (Grade 440c) used in friction tests (model 312-TBR-T, 7.94 mm in diameter) were purchased from Bal-Tec (USA) and cleaned in the same way as the stainless steel sheets before use. The surface roughness of the steel balls, measured using AFM, was $2.2 \pm 0.3\ nm$ at a scan scale of 5 μm by 5 μm [99].

2.6. AFM experiments

A Veeco Nanoscope IV AFM was used to measure normal force vs separation curves and nanoscale frictional (lateral) force vs normal load curves on a freshly cleaned Grade 304 stainless steel substrate. Sharp silicon AFM tips (NSC36, Mikromasch) with spring constants of $0.7 \pm 0.3\ N\ m^{-1}$ were cleaned with ethanol and MilliQ water, followed by UV-Ozone for 5 min before use. An AFM fluid cell (Bruker) was washed with ethanol and MilliQ water, then dried with nitrogen flow before each experiment.

Normal force vs separation curves were collected by moving the surface towards the AFM tip and detecting the cantilever deflection as a function of separation. The ramp size ranged from 50 nm to 100 nm and the ramp rate was 0.2 Hz. Standard methods were used to convert deflection vs separation data to normal force vs separation curves [100]. Lateral forces were obtained by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled. The scan size was 500 nm, and scan rate was 6 Hz. The lateral deflection signal (i.e., cantilever twist) was converted to lateral force using a customized function produced in MATLAB, which takes into account the torsional spring constant and the geometrical dimensions of the cantilever [101].

The surface potential was controlled by a WaveNano® potentiostat with the stainless steel surface as the working electrode, and platinum wires (PT5441, diameter 0.40 mm, purity 99.99 %, Advent Research Materials) as both counter and reference electrodes. The configuration of electrodes for AFM experiments are shown in Figure S3. Every experiment was repeated at least twice.

3. Results and discussion

3.1. Miscibility

The chemical structures of the SAIL ions, HD, and (CH₂CO₂Et)₂ are presented in Table 1. Fig. 1 shows the miscibility of [P_{6,6,6,14}] [AOT] and [P_{6,6,6,14}] [DS] in HD and (CH₂CO₂Et)₂. [P_{6,6,6,14}] [AOT] is fully miscible with HD and with (CH₂CO₂Et)₂ up to 45 wt%. This is because [P_{6,6,6,14}]⁺ is a well-known hydrophobic cation [102–104], while [AOT][−] ions tend to be miscible with apolar solvents but has limited miscibility with water [105–108]. Therefore, it is reasonable that [P_{6,6,6,14}] [AOT] is fully miscible with the apolar HD, but only partially miscible with the polar (CH₂CO₂Et)₂. In contrast, while [P_{6,6,6,14}] [DS] is fully miscible with (CH₂CO₂Et)₂, it is only miscible up to only 2 wt% in HD due to the limited solubility of [DS][−] ions in apolar oils caused by the exposure of its polar charged groups [109,110].

3.2. Macroscale friction

The thickness of the lubricant film plays a critical role in determining the lubrication regime. Table 2 lists the film thicknesses (*h*) of [P_{6,6,6,14}] [AOT] and [P_{6,6,6,14}] [DS] in HD or (CH₂CO₂Et)₂, based on the experimental viscosities (*η*), a pressure-viscosity coefficient (*α*) of 12 GPa^{−1}, and the Hamrock and Dowson model [37,113,114]. The *α* used in these calculations is based on the literature [58, 59, 115–119], as detailed in the SI. The effect of varying *α* is presented in Table S1 and shows that while the thickness values may vary, each sample remains in the same lubrication regime and therefore the logical conclusions are the same. Further details of the film thickness calculations are also presented in the

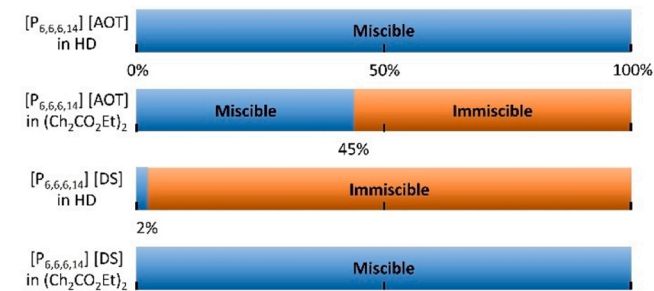


Fig. 1. Miscibility of [P_{6,6,6,14}] [AOT] or [P_{6,6,6,14}] [DS] in HD or (CH₂CO₂Et)₂ at room temperature.

SI. The surface roughness for the stainless steel sheets and ball used is 0.9 nm and 2.2 nm respectively, which results in a composite surface roughness of 2.4 nm. The ratio of the estimated *h* to the composite surface roughness is represented by *λ*. A *λ* < 1.2 is considered to be in the boundary lubrication regime, 1.2 < *λ* < 3.0 is in mixed lubrication regime, 3.0 < *λ* < 10.0 is in elastohydrodynamic lubrication regime, and *λ* > 10.0 is in hydrodynamic lubrication regime [99,120]. Table 2 shows that both pure [P_{6,6,6,14}] [AOT] and pure [P_{6,6,6,14}] [DS] are in the hydrodynamic lubrication regime, 80 wt% [P_{6,6,6,14}] [DS] in (CH₂CO₂Et)₂ is in the elastohydrodynamic lubrication regime, and 50 wt% [P_{6,6,6,14}] [DS] in (CH₂CO₂Et)₂ is in the mixed lubrication regime. All the other mixtures and both pure HD and pure (CH₂CO₂Et)₂ are in the boundary lubrication regime.

Fig. 2A–D show macroscale plots of the friction coefficient (*μ*), which

Table 1
Dimensions of [P_{6,6,6,14}]⁺, [AOT][−], and [DS][−] ions, and HD and (CH₂CO₂Et)₂ molecules, calculated using Avogadro software [111,112].

Ion/Molecule Name	Abbreviation	Structure
trihexyltetradecylphosphonium	[P _{6,6,6,14}] ⁺	
dioctyl sulfosuccinate	[AOT] [−]	
dodecyl sulfate	[DS] [−]	
hexadecane	HD	
diethyl succinate	(CH ₂ CO ₂ Et) ₂	

Table 2

Experimental viscosities (η), calculated film thicknesses (h), estimated ratios of film thickness and surface roughness (λ), and average friction coefficients (μ) of [P_{6,6,6,14}] [AOT] and [P_{6,6,6,14}] [DS] in HD and (CH₂CO₂Et)₂ at the macroscale. The η values were collected at the temperature of 25 °C (298 K) at a fixed shear rate of 50 s⁻¹. The errors are within 10 %. The h values were based on the η values for the pressure-viscosity coefficient (α) of 12 GPa⁻¹. The λ values were based on the h values and the surface roughness of 2.4 nm from AFM measurements. The μ values are extracted from Fig. 2.

Sample	η (mPa.s)	h (nm)	λ	μ
Pure HD	2.05	0.97	0.40	0.15
Pure (CH ₂ CO ₂ Et) ₂	0.91	0.56	0.23	0.35
2 wt% [P _{6,6,6,14}] [AOT] in HD	1.81	0.89	0.37	0.15
5 wt% [P _{6,6,6,14}] [AOT] in HD	2.32	1.05	0.44	0.15
2 wt% [P _{6,6,6,14}] [AOT] in (CH ₂ CO ₂ Et) ₂	0.98	0.58	0.24	0.21
5 wt% [P _{6,6,6,14}] [AOT] in (CH ₂ CO ₂ Et) ₂	1.26	0.69	0.29	0.21
25 wt% [P _{6,6,6,14}] [AOT] in (CH ₂ CO ₂ Et) ₂	2.98	1.25	0.52	0.19
2 wt% [P _{6,6,6,14}] [DS] in HD	2.16	1.00	0.42	0.11
2 wt% [P _{6,6,6,14}] [DS] in (CH ₂ CO ₂ Et) ₂	1.66	0.84	0.35	0.16
5 wt% [P _{6,6,6,14}] [DS] in (CH ₂ CO ₂ Et) ₂	2.76	1.18	0.49	0.15
25 wt% [P _{6,6,6,14}] [DS] in (CH ₂ CO ₂ Et) ₂	3.56	1.41	0.59	0.15
50 wt% [P _{6,6,6,14}] [DS] in (CH ₂ CO ₂ Et) ₂	11.8	3.18	1.33	0.12
80 wt% [P _{6,6,6,14}] [DS] in (CH ₂ CO ₂ Et) ₂	52.2	8.72	3.63	0.09
Pure [P _{6,6,6,14}] [AOT]	1239	75.25	31.3	0.15
Pure [P _{6,6,6,14}] [DS]	1143	71.20	29.7	0.05

is the proportionality between the lateral force and the normal force, versus sliding distance for stainless steel surfaces immersed in the pure SAILs [P_{6,6,6,14}] [AOT] and [P_{6,6,6,14}] [DS], the pure base liquids HD and

(CH₂CO₂Et)₂, and several concentrations of each SAIL in each base liquid. The complete set of individual macroscale friction plots can be found in Fig. S4-S6 of the SI.

Fig. 2 shows that for most samples, μ varies considerably before reaching a stable value, which is known as the “running-in period” [38, 121]. For the stable values, μ decreases as follows: pure base liquids > SAIL solutions > pure SAILs, except for [P_{6,6,6,14}] [AOT] in HD which have the same friction (Fig. 2B). For the pure base liquids, μ of (CH₂CO₂Et)₂ (0.35) is much higher than HD (0.15), despite similar viscosity (Table S1) and both liquids being boundary lubricating. This is attributed to the longer alkyl chain of HD, which separates surfaces more effectively than the shorter chains of (CH₂CO₂Et)₂, resulting in better macroscale lubricity for HD as supported by previous studies [122–124].

The pure SAILs have low friction at the macroscale, but μ is higher for [P_{6,6,6,14}] [AOT] (0.15) than for [P_{6,6,6,14}] [DS] (0.05). The Hamrock and Dowson model shows both pure SAILs are in the hydrodynamic regime, so μ is proportional to the viscosity of the lubricant [38]. At room temperature, [P_{6,6,6,14}] [AOT] has a higher viscosity (1239 mPa.s) compared to [P_{6,6,6,14}] [DS] (1143 mPa.s), c.f. Table S1, which is consistent with previous studies [125–127]. However, the difference in μ is much greater than the difference in viscosity. This is likely due to the viscosity of the bulk IL differing from the viscosity of the confined IL, as previously observed [111].

For SAIL solutions, μ is typically close to the pure SAIL and significantly lower than the pure base liquids. μ increases as SAIL concentration decreases, consistent with macroscale friction studies of conventional ILs as additives [128]. All SAIL solutions are in the

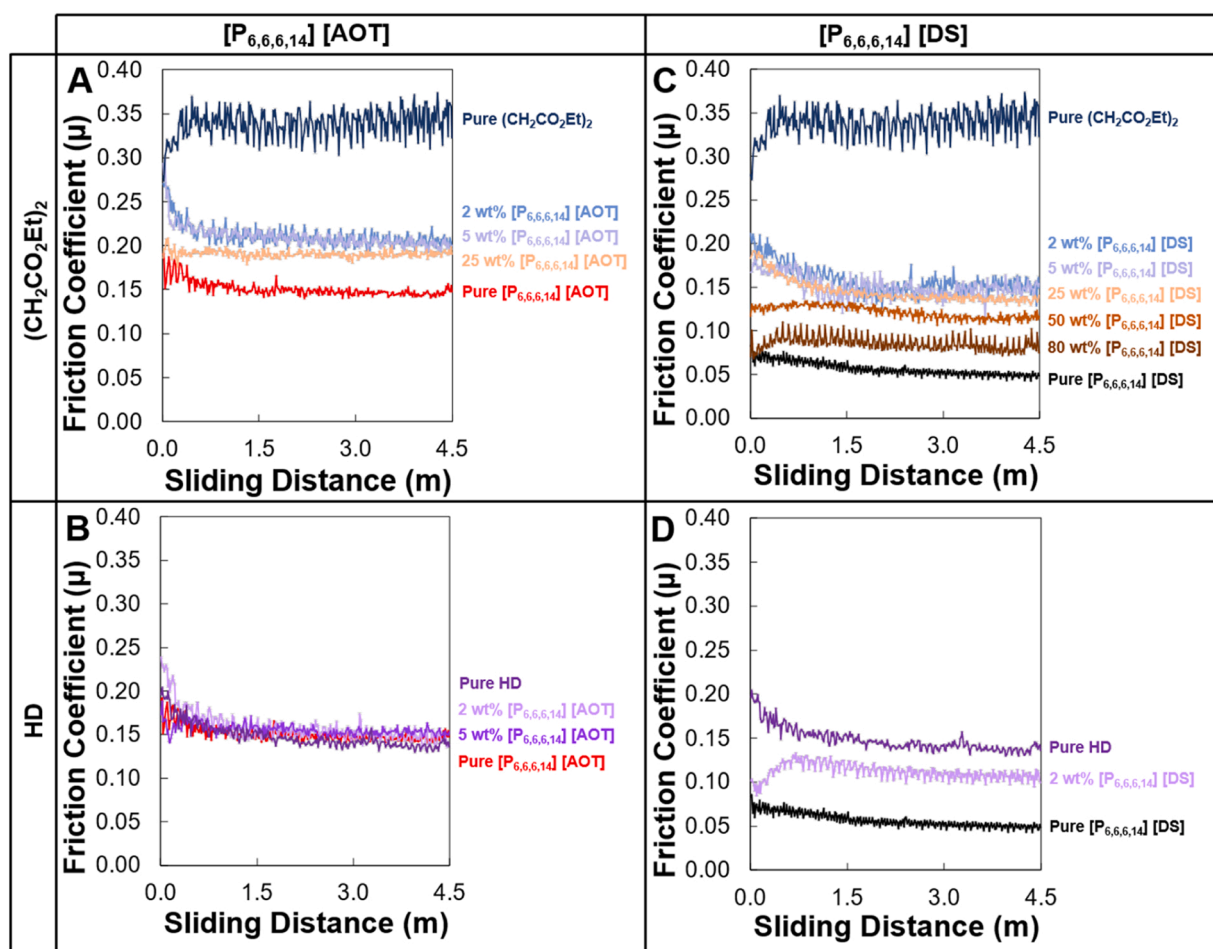


Fig. 2. Friction coefficients (μ) vs sliding distance of stainless steel balls on a stainless steel disc immersed in (A) [P_{6,6,6,14}] [AOT] in (CH₂CO₂Et)₂, (B) [P_{6,6,6,14}] [AOT] in HD, (C) [P_{6,6,6,14}] [DS] in (CH₂CO₂Et)₂, and (D) [P_{6,6,6,14}] [DS] in HD. Data points are the averaged results of duplicate runs. Error bars, which are smaller than the data points, show the standard error of each point.

boundary lubrication regime, except for 80 wt% and 50 wt% $[P_{6,6,6,14}]$ [DS] in $(CH_2CO_2Et)_2$ which are in the elastohydrodynamic regime and mixed regime respectively, therefore differences in μ are due to surface excesses of SAIL ions adsorbed onto the stainless steel surface. Higher concentrations of a SAIL in a base liquid significantly increase enrichment of SAIL ions in the boundary layer, strengthening the interfacial structure and improving lubrication [128].

$[P_{6,6,6,14}]$ [AOT] in HD is an exception, as pure $[P_{6,6,6,14}]$ [AOT], pure HD, and their solutions have the same μ (Fig. 2B). These findings initially seem counterintuitive but can be attributed to different lubrication regimes. Pure $[P_{6,6,6,14}]$ [AOT] is in the hydrodynamic lubrication regime, where high viscosity leads to high friction. In contrast, both pure HD and $[P_{6,6,6,14}]$ [AOT] in HD are in the boundary lubrication regime, where the long alkyl chain of HD improves lubrication compared to shorter chain of $(CH_2CO_2Et)_2$, compensating for the viscosity differences. As a result, these liquids display similar friction values at the macroscale, despite their distinct lubrication mechanisms.

3.3. Interfacial nanostructure and friction at the nanoscale

Nanoscale friction investigated using AFM is vital for a more profound understanding of the lubrication mechanisms, as **it is less affected by surface asperities than macroscale measurements**. The maximum

contact pressure between the AFM tip and the surface is **0.45 GPa**, although slightly lower than that in the macroscale friction tests (0.67 GPa), as detailed in the SI, the sliding velocity at the nanoscale (**$6 \mu m s^{-1}$**) is more than three orders of magnitude smaller than that in the macroscale friction tests ($15 mm s^{-1}$). Therefore, all samples fall under the boundary lubrication regime and the shear plane is on the boundary layer. Consequently, the interfacial nanostructure, especially the boundary layer, has a significant effect on friction at the nanoscale.

AFM normal force-separation profiles were used to study the interfacial nanostructures of pure SAILs, pure base liquids, and 2 wt% and 5 wt% of each SAIL in each base liquid at stainless steel surfaces (Fig. 3A-B). Further representative force-separation profiles, including both approach and retraction data for all samples, are listed in Fig. S7-S9 in the SI. The profiles reveal the key features of several hundred force curves collected over different experiments. A single representative force curve is shown instead of an average, as averaging can mask key features at low normal forces [17,129]. In an AFM experiment, the absolute distance between the AFM tip and the surface cannot be ascertained, as the compliance region at zero apparent separation corresponds to the AFM tip pushing against material that cannot be displaced, which is not necessarily surface contact. As the AFM tip approaches the surface from a large apparent separation, the measured force is zero.

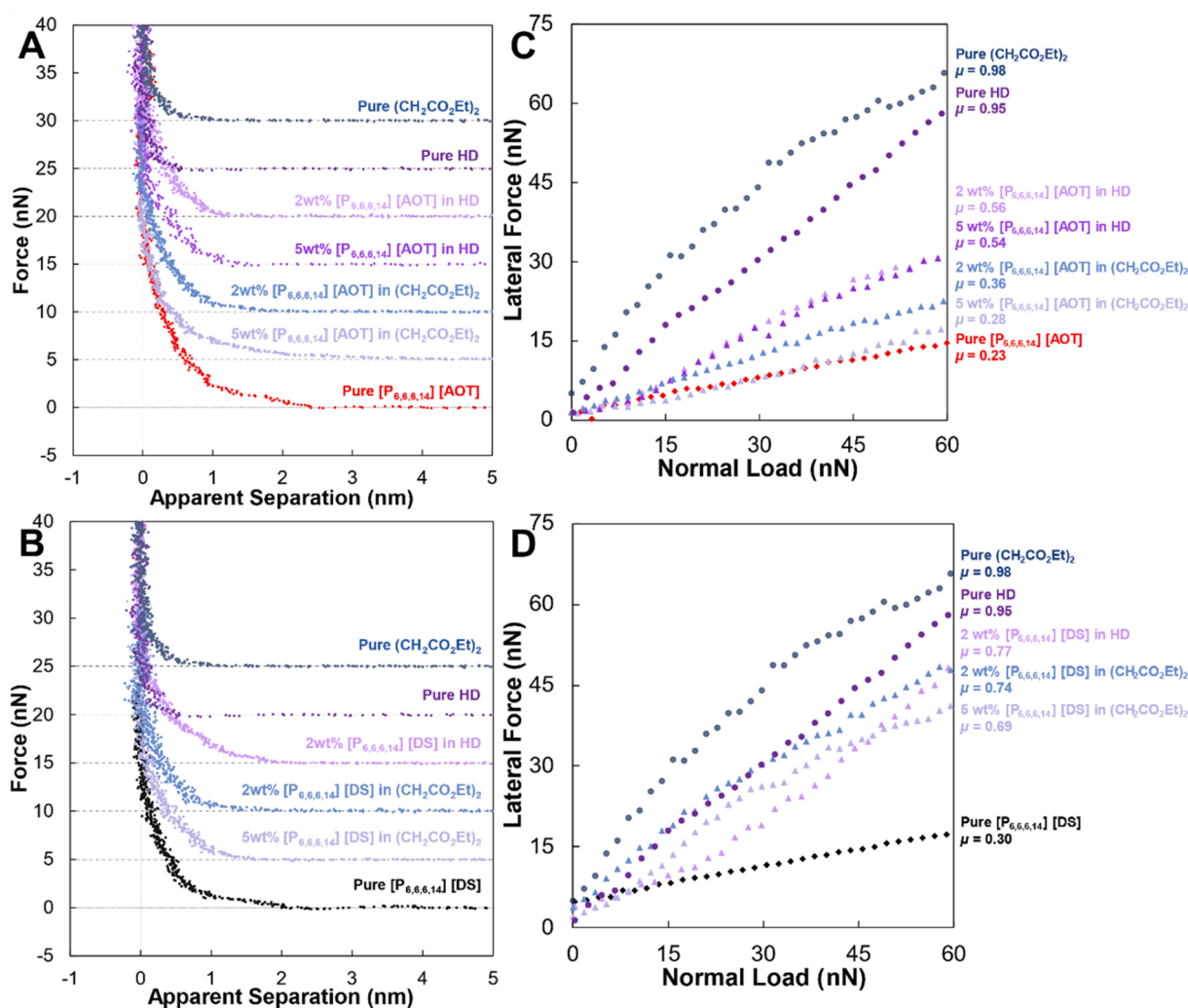


Fig. 3. Representative normal force-separation profiles of a silicon AFM tip approaching stainless steel immersed in (A) pure $[P_{6,6,6,14}]$ [AOT], pure HD, pure $(CH_2CO_2Et)_2$, and their solutions, and (B) pure $[P_{6,6,6,14}]$ [DS], pure HD, pure $(CH_2CO_2Et)_2$, and their solutions. Lateral force vs normal force of a silicon AFM tip on stainless steel immersed in (C) pure $[P_{6,6,6,14}]$ [AOT], pure HD, pure $(CH_2CO_2Et)_2$, and their solutions, and (D) pure $[P_{6,6,6,14}]$ [DS], pure HD, pure $(CH_2CO_2Et)_2$, and their solutions. The errors are within 10 %.

Recent AFM video-rate imaging studies on ILs [44] have revealed that amphiphilically nanostructured ILs exhibit a distinctive near-surface nanostructure, where the upper half of a near-surface aggregate diffuses slowly over the boundary layer. As the SAILs investigated are also amphiphilically nanostructured [24,30], the near-surface structure is expected to be similar. The number of steps in the force curves and the push-through forces tend to increase with the degree of interfacial liquid structure [17,130,131].

In Fig. 3A–B, at an apparent separation of 2.3 nm for pure $[P_{6,6,6,14}]$ [AOT] and 2.0 nm for pure $[P_{6,6,6,14}]$ [DS], a repulsive force is detected, which indicates that the AFM tip is in contact with the near-surface structure. This repulsive force increases monotonically to 2 nN at an apparent separation of 1.0 nm for pure $[P_{6,6,6,14}]$ [AOT] and 0.7 nm for pure $[P_{6,6,6,14}]$ [DS]. The step sizes are much smaller than the ion/ion pair dimensions listed in Table 1, which means the AFM tip has compressed the near-surface structure. A gentle push-through to compliance follows, where the AFM tip is on the boundary layer that it cannot displace [17]. The pure SAIL force curves are not as distinct as those seen with other ILs such as ethylammonium nitrate, due to the rougher stainless steel surface smoothing out the interfacial structure and the greater compressibility of the SAIL alkyl chains being squeezed by the AFM tip reducing the step sizes [132,133]. Previous studies found $[P_{6,6,6,14}]$ [AOT] has a step size of 1.7 nm at both graphite and gold surfaces which are slightly negatively charged at the open circuit potential (OCP), likely due to interdigitated ion pairs [24]. Here, the force curves shown in Fig. 3A indicate that $[P_{6,6,6,14}]$ [AOT] has a step size of 1.0 nm at stainless steel surfaces. **The stainless steel surface has an average OCP value of +0.24 V in the pure SAILs**, so the boundary layer is anion-enriched, as observed in previous studies [24,30]. The associated alkyl chains are orientated away from the surface and form a well-defined interfacial structure [80,134] (Fig. 4). It is difficult for the AFM tip to push through the boundary layer [17], and thus it is likely the final push-through layer found here (~1.0 nm) is enriched in $[P_{6,6,6,14}]^+$ cations, which are compressed as the AFM tip moves closer to the surface.

The interfacial structures of the pure base liquids are significantly less defined than the pure SAILs and the SAIL solutions. Pure HD and pure $(CH_2CO_2Et)_2$ show repulsion starting between 0.5–0.7 nm and immediately enters a push-through to compliance, which is typical of electrostatic repulsion [73,79]. All diluted SAIL solutions show repulsion starting between 1.0–2.0 nm and increase monotonically to 1 nN–2 nN before a gentle push-through to compliance, consistent with the pure SAILs. The push-through steps are less defined with the addition of HD or $(CH_2CO_2Et)_2$, indicating weaker interfacial structure. However, the consistent compressions indicate that the composition of the boundary layer remains enriched in SAIL anion head groups. Any small differences can be attributed to the rougher stainless steel surface flattening out the interfacial structure compared to the smoother graphite and gold surfaces.

Fig. 3C–D present nanofriction (lateral) force versus normal force of stainless steel immersed in $[P_{6,6,6,14}]$ [AOT], $[P_{6,6,6,14}]$ [DS], HD, $(CH_2CO_2Et)_2$, and several concentrations of each SAIL in each base liquid. The friction coefficient (μ) for each condition is extracted from the gradient of these plots and is also shown in Fig. 3C–D.

Fig. 3C shows that μ decreases as follows: pure HD > pure $(CH_2CO_2Et)_2$ > $[P_{6,6,6,14}]$ [AOT] in HD > $[P_{6,6,6,14}]$ [AOT] in $(CH_2CO_2Et)_2$ > pure $[P_{6,6,6,14}]$ [AOT]. Fig. 3D shows similar results for $[P_{6,6,6,14}]$ [DS] in HD and in $(CH_2CO_2Et)_2$, except for a smaller decrease in μ from the pure base liquids to the $[P_{6,6,6,14}]$ [DS] solutions in base liquids at higher normal loads. Both figures show results consistent with macroscale friction in Fig. 2 and previous nanofriction studies [24,30,59,79,135,136]. However, clear differences are seen for pure $[P_{6,6,6,14}]$ [AOT], pure $[P_{6,6,6,14}]$ [DS], and their solutions in base liquids.

At the nanoscale, the lower friction for pure HD compared to pure $(CH_2CO_2Et)_2$ is consistent with the trend seen at the macroscale, confirming that long alkyl chains assist the formation of a more robust

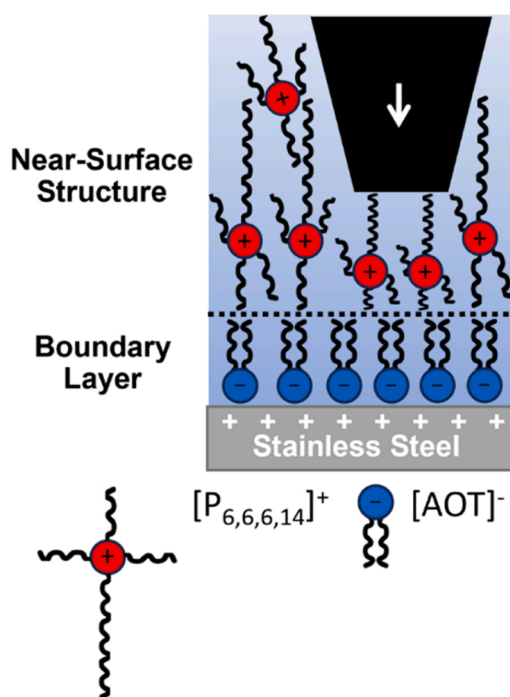


Fig. 4. Basic schematic illustrating the interfacial structure of stainless steel immersed in $[P_{6,6,6,14}]$ [AOT] at the nanoscale. Note that the actual structure is highly dynamic and more complex than that depicted in the schematic.

boundary layer to resist friction in the boundary lubrication regime at both the nanoscale and the macroscale. However, pure $[P_{6,6,6,14}]$ [AOT] has a lower μ than pure $[P_{6,6,6,14}]$ [AOT] (Fig. 3), which is opposite to the behaviour observed at the macroscale. This discrepancy likely arises from differences in lubrication regimes, as the pure SAILs are in the boundary lubrication regime during nanoscale friction measurements, but in the hydrodynamic lubrication regime during macroscale friction tests. The higher viscosity of $[P_{6,6,6,14}]$ [AOT] compared to $[P_{6,6,6,14}]$ [DS] in the hydrodynamic regime, as well as the viscosity of the bulk IL differing from the viscosity of the confined IL, as previously observed [111], lead to higher friction at the macroscale. However, friction in the boundary regime is mainly attributed to the robustness of the boundary layers adsorbed onto the sliding surfaces [38], and is thus independent of lubricant viscosity. When confined at the nanoscale, the long alkyl chain of $[DS]^-$ ions sterically hinder other $[DS]^-$ ions from adsorbing strongly onto the surface and restrict packing compared to the more branched $[AOT]^-$ ions [38]. The bulkier structure of $[AOT]^-$ ions produces a stronger near-surface structure than $[DS]^-$ ions [22,137,138]. Therefore, the boundary layer formed by $[P_{6,6,6,14}]$ [AOT] is more effective at minimizing surface contact in the boundary regime, resulting in lower friction at the nanoscale.

The nanoscale friction coefficients of SAILs in base liquids fall between those of pure SAILs and pure base liquids. Fig. 3C shows that both 2 wt% and 5 wt% $[P_{6,6,6,14}]$ [AOT] in $(CH_2CO_2Et)_2$ exhibit low friction which is comparable to pure $[P_{6,6,6,14}]$ [AOT] across the entire range of normal loads investigated. Notably, increasing the concentration of $[P_{6,6,6,14}]$ [AOT] results in further decreases of friction, which is likely due to a higher surface excess of $[AOT]^-$ ions at higher $[P_{6,6,6,14}]$ [AOT] concentrations.

For $[P_{6,6,6,14}]$ [AOT] in HD at concentrations of 2 wt% and 5 wt%, the nanoscale friction coefficients are almost identical to each other and as low as those of pure $[P_{6,6,6,14}]$ [AOT] and $[P_{6,6,6,14}]$ [AOT] in $(CH_2CO_2Et)_2$ up to a normal load of 15 nN. However, at higher normal loads, the friction coefficients of $[P_{6,6,6,14}]$ [AOT] in HD increase. The increase in friction coefficient at higher normal loads, which is a non-linear relationship between lateral force and normal load, is typically

indicative of the penetration of an adsorbed ion layer, as previously observed [128]. The increase in friction coefficient with normal force is observed for $[P_{6,6,6,14}] [AOT]$ in HD but not for $[P_{6,6,6,14}] [AOT]$ in $(CH_2CO_2Et)_2$. This suggests that the boundary layer formed by $[P_{6,6,6,14}] [AOT]$ in HD is weaker and can be disrupted at higher normal forces, which is consistent with previous miscibility studies that show that $[P_{6,6,6,14}] [AOT]$ is more miscible with HD than $(CH_2CO_2Et)_2$. Although 2 wt% and 5 wt% $[P_{6,6,6,14}] [AOT]$ form homogenous solutions in both HD and $(CH_2CO_2Et)_2$, the surface excess of $[P_{6,6,6,14}] [AOT]$ at the stainless steel surface is likely lower when dissolved in HD due to its higher miscibility in HD compared to $(CH_2CO_2Et)_2$. This lower surface excess results in the formation of weaker boundary layers. At very high normal loads the friction coefficient of $[P_{6,6,6,14}] [AOT]$ in HD decrease to values comparable to that observed at low normal loads (<15 nN). Although the reason for this is unclear, this behaviour might be attributed to the kinetics of re-adsorption during the friction measurement. It is plausible that the displaced $[AOT]^-$ ions are able to reform a boundary layer with comparable robustness to that observed at lower loads, similar as seen before [55].

For $[P_{6,6,6,14}] [DS]$ -based solutions (Figs. 3D), 2 wt% $[P_{6,6,6,14}] [DS]$ in HD exhibits comparably low nanoscale friction to pure $[P_{6,6,6,14}] [DS]$ up to the normal load of 20 nN. However, at higher normal loads, the friction increases significantly and approaches that of pure HD. This behaviour indicates that a boundary layer enriched in $[P_{6,6,6,14}] [DS]$ forms at the stainless steel surface, but is disrupted when the normal load exceeds 20 nN. The friction of higher concentrations of $[P_{6,6,6,14}] [DS]$ in HD was not investigated due to the limited miscibility of the SAIL in the base liquid. Interestingly, 2 wt% and 5 wt% $[P_{6,6,6,14}] [DS]$ in $(CH_2CO_2Et)_2$ exhibit even worse lubricity than 2 wt% $[P_{6,6,6,14}] [DS]$ in HD. The friction coefficients of these solutions rise sharply with normal load and do not show any obvious increase, decrease, or flattening at higher normal loads. This difference in behaviour may be attributed to the fact that $[P_{6,6,6,14}] [DS]$ is fully miscible in $(CH_2CO_2Et)_2$, but only up to 2 wt% in HD. Consequently, at the same concentration, the surface excess of $[P_{6,6,6,14}] [DS]$ at the stainless steel surface is likely higher in HD than in $(CH_2CO_2Et)_2$, resulting in better lubricity. This observation is consistent with the observed results for $[P_{6,6,6,14}] [AOT]$ -based solutions.

These findings indicate that using SAILS as lubricant additives in base liquids may not be advantageous for boundary lubrication, as the surface excess of the SAILS at the interface might be insufficient, though the high miscibility of SAILS in base liquids results in stable solutions. Therefore, it is crucial to find an optimal balance between the miscibility of SAILS and their surface excess at the interface to achieve the best lubricity.

3.4. Tribotronics at the nanoscale

Fig. 5A-B show nanoscale friction (lateral) force as a function of normal force of stainless steel immersed in pure $[P_{6,6,6,14}] [AOT]$ and 5 wt% $[P_{6,6,6,14}] [AOT]$ in $(CH_2CO_2Et)_2$ at various potentials. The friction coefficient for each condition, extracted from the gradient of these plots, is also shown in Fig. 5A-B.

For pure $[P_{6,6,6,14}] [AOT]$, friction is highest at OCP -1.0 V, while friction at the OCP and OCP $+1.0$ V is much lower and approximately equal (Fig. 5A). In the case of 5 wt% $[P_{6,6,6,14}] [AOT]$ in $(CH_2CO_2Et)_2$, friction coefficients are generally higher, but the trend is similar. Friction decreases as follows: OCP -1.0 V $>$ OCP $>$ OCP $+1.0$ V (Fig. 5B). However, in this system, friction at the OCP is higher and comparable to that at OCP -1.0 V. For both pure $[P_{6,6,6,14}] [AOT]$ and 5 wt% $[P_{6,6,6,14}] [AOT]$ in $(CH_2CO_2Et)_2$, the slightly positive OCP ($+0.24$ V) attracts $[AOT]^-$ ions to the stainless steel surface, where they adsorb and form a robust boundary layer [52]. The presence of $(CH_2CO_2Et)_2$ molecules in the 5 wt% $[P_{6,6,6,14}] [AOT]$ solution hinders the same level of anion enrichment in the boundary layer compared to pure $[P_{6,6,6,14}] [AOT]$, resulting in a higher μ at the OCP.

At OCP $+1.0$ V, the increased electrostatic interactions between the

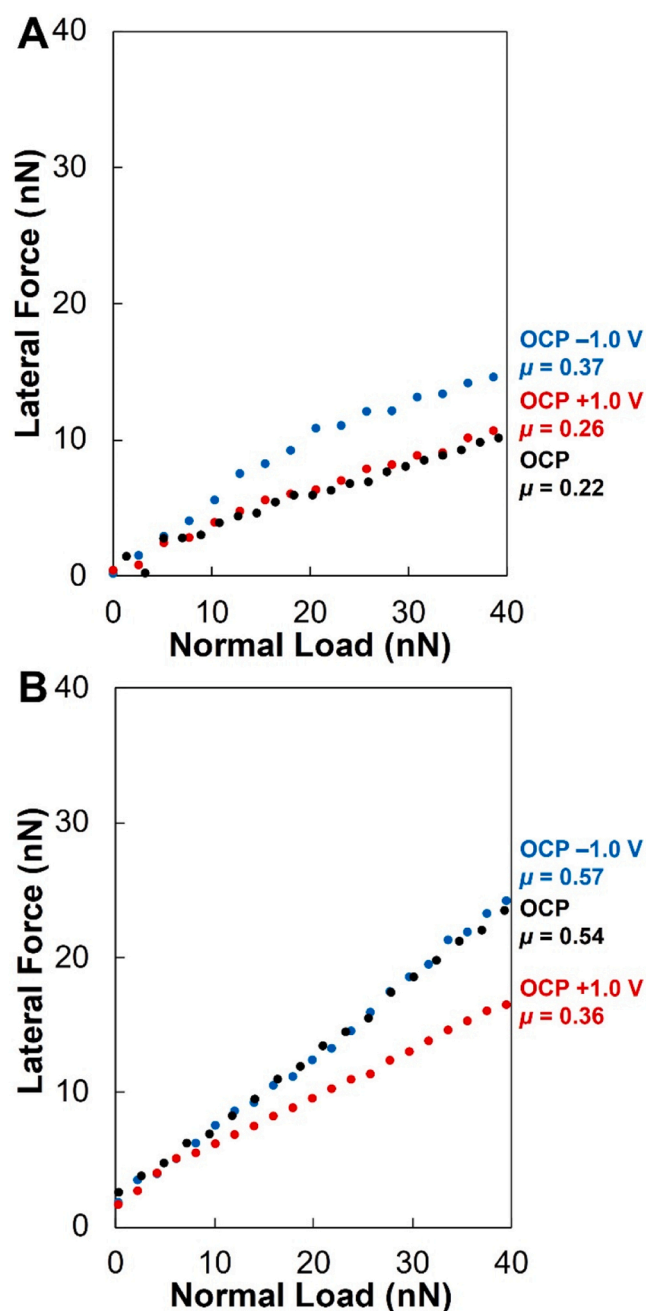


Fig. 5. Lateral force vs normal load of a silicon AFM tip approaching stainless steel immersed in (A) pure $[P_{6,6,6,14}] [AOT]$ and (B) 5 wt% $[P_{6,6,6,14}] [AOT]$ in $(CH_2CO_2Et)_2$ at the OCP, OCP -1.0 V, and OCP $+1.0$ V. The errors are within 10 %.

surface and $[AOT]^-$ ions leads to a higher anion enrichment at the interface, forming a stronger boundary layer that improves lubrication. Remarkably, this effect is more pronounced for 5 wt% $[P_{6,6,6,14}] [AOT]$ in $(CH_2CO_2Et)_2$ than for pure $[P_{6,6,6,14}] [AOT]$. This difference may be because the boundary layer in pure $[P_{6,6,6,14}] [AOT]$ is already saturated with $[AOT]^-$ ions at the OCP, or because the incorporation of additional $[AOT]^-$ ions into the boundary layer does not significantly improve its robustness. At OCP -1.0 V, the negatively charged surface attracts $[P_{6,6,6,14}]^+$ ions while repelling $[AOT]^-$ ions, displacing them from the boundary layer. The higher μ at OCP -1.0 V compared to OCP $+1.0$ V indicates that $[P_{6,6,6,14}]^+$ ions are less lubricating than $[AOT]^-$ ions on stainless steel at the nanoscale. This observation differs from previous studies on gold and graphite, where $[P_{6,6,6,14}]^+$ ions at negative

potentials exhibit better lubricity than [AOT]⁻ ions at positive potentials [24]. This discrepancy may be due to the much higher surface roughness of stainless steel compared to the atomically smooth surfaces of gold and graphite, which results in weaker boundary layers enriched in [P_{6,6,6,14}]⁺ ions, or due to a change in energy dissipation from differences in ion structure, conformation or packing density in the near-surface layers [39,40].

4. Conclusion

Two SAILs, [P_{6,6,6,14}] [AOT] and [P_{6,6,6,14}] [DS], have been explored for the first time as lubricant additives in both polar and apolar base liquids. Macroscale and nanoscale friction results demonstrate that the addition of small quantities of these SAILs to base liquids significantly enhances lubrication on stainless steel surfaces. The enhanced lubrication is primarily attributed to the adsorption of SAIL anions onto the positively charged stainless steel surface, which leads to the formation of a robust boundary layer that prevents surface contact and facilitates smooth sliding. The friction coefficients of the pure SAILs are up to ~80 % lower than those of pure base liquids. Remarkably, SAIL solutions diluted to just 5 wt% exhibit lubrication performance comparable to that of pure SAILs, indicating their great potential as lubricant additives. At the macroscale [P_{6,6,6,14}] [DS] shows a better lubrication performance in both base liquids compared to [P_{6,6,6,14}] [AOT], whereas the opposite is true at the nanoscale. This is explained by differences in the miscibility of each SAIL in polar (CH₂CO₂Et)₂ or apolar HD, and differences in lubrication regimes between macroscale and nanoscale friction. AFM force-separation profiles confirm that diluted solutions form interfacial structures similar to those of pure SAILs. Tribotronic studies reveal that on stainless steel surfaces, the anion-rich boundary layer formed at OCP +1.0 V provides better lubrication compared to the cation-rich boundary layer formed at OCP -1.0 V. These findings contribute to the advancement of high-performance and cost-effective lubricant additives, with potential implications for various industrial sectors that require durable and reliable lubrication.

CRediT authorship contribution statement

Joshua J. Buzolic: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Hua Li:** Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization. **Zachary M. Aman:** Supervision, Resources. **Debbie S. Silvester:** Supervision, Resources. **Rob Atkin:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude in order to improve language. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.colsurfa.2024.134669.

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